

# IMO (57<sup>th</sup>) – 2016

## First Day

1. Triangle  $BCF$  has a right angle at  $B$ . Let  $A$  be the point on line  $CF$  such that  $FA = FB$  and  $F$  lies between  $A$  and  $C$ . Point  $D$  is chosen such that  $DA = DC$  and  $AC$  is the bisector of  $\angle DAB$ . Point  $E$  is chosen such that  $EA = ED$  and  $AD$  is the bisector of  $\angle EAC$ . Let  $M$  be the midpoint of  $CF$ . Let  $X$  be the point such that  $AMXE$  is a parallelogram (where  $AM \parallel EX$  and  $AE \parallel MX$ ). Prove that lines  $BD$ ,  $FX$  and  $ME$  are concurrent.
2. Find all positive integers  $n$  for which each cell of an  $n \times n$  table be filled with one of the letters  $I, M, O$  in such a way that:
  - (a) In each row and each column, one third of the entries are  $I$ , one third are  $M$  and one third are  $O$ ; and
  - (b) In any diagonal, if the number of entries on the diagonal is a multiple of three, then one third of the entries are  $I$ , one third are  $M$  and one third are  $O$ .

Note: The row and column of an  $n \times n$  table are each labeled 1 to  $n$  in a natural order. Thus each cell corresponds to a pair of positive integers  $(i, j)$  with  $1 \leq i, j \leq n$ . For  $n > 1$ , the table has  $4n - 2$  diagonals of two types. A diagonal of the first type consists of all cells  $(i, j)$  for which  $i + j$  is a constant, and a diagonal of second type consists of all cells  $(i, j)$  for which  $i - j$  is a constant.

3. Let  $P = A_1 A_2 \dots A_k$  be a convex polygon in the plane. The vertices  $A_1, A_2, \dots, A_k$  have integral coordinates and lie on a circle. Let  $S$  be the area of  $P$ . An odd positive integer  $n$  is given such that the square of the side lengths  $P$  are integer divisible by  $n$ . Prove that  $2S$  is an integer divisible by  $n$ .

## Second Day

4. A set of positive integers is called as Fragrant if it contains at least two elements and each of its elements has a prime factor with at least one of the other elements. Let  $P(n) = n^2 + n + 1$ . What is the least possible value of the positive integer  $b$  such that there exist a non-negative integer  $a$  for which the set  $\{P(a+1), P(a+2), \dots, P(a+b)\}$  is fragrant?

## IMO QUESTIONS

5. The Equation  $(x-1)(x-2)\dots(x-2016) = (x-1)(x-2)\dots(x-2016)$  is written on the board, with 2016 linear factors on each side. What is the least positive value of  $k$  for which it is possible to erase exactly  $k$  of these 4032 linear factors so that at least one factor remains on each side and the resulting equation has no real solution?
6. There are  $n \geq 2$  line segments in the plane such that every two segments cross and no three segments meet at a point. Geoff has to choose an endpoint of each segment and place a frog on it, facing the other endpoint. Then he will clap his hands  $n-1$  times. Every time he claps, each frog will immediately jump forward to the next intersection point on its segment. Frog never changes the direction of their jump. Geoff wishes to place the frog in such a way that no two of them will ever occupy the same intersection point at the same time.
- (a) Prove that Geoff can always fulfill his wish if  $n$  is odd.
- (b) Prove that Geoff can always fulfill his wish if  $n$  is even.

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